Status	Finished
Started	Friday, 16 October 2026, 12:37 PM
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Duration	3 mins 1 sec
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- a. 90 and 99 ✓
- b. 90 and 94
- c. 89 and 99
- d. 89 and 94

Question **2**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Linear
- b. Hash
- c. Both (1) & (3)
- d. Binary ✓

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is very small
- b. When the list contains duplicate elements
- c. When the list is unsorted
- d. When the list is sorted ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search can be applied to both sorted and unsorted lists. ✓
- b. Linear search divides the list into halves.
- c. Linear search is more efficient than binary search on large lists.
- d. Linear search requires the list to be sorted.

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. There must be a mechanism to delete and/or insert elements in the list ✓
- b. Number values should only be present

  Number values should only be present
- c. There should be direct access to the middle element in any sublist
- d. The list must be sorted

Question **6**

Correct

Mark 1.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The search stops
- b. The higher middle element is chosen as the middle element ✓
- c. The middle element is chosen randomly
- d. The lower middle element is chosen as the middle element

Question **7**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Hashing
- b. Jump search
- c. Binary search
- d. Linear search ✓

Question **8**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The first element is returned
b. The last element is returned
c. The search is considered unsuccessful ✓
d. An error is raised

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Dynamic search
b. Hashing
c. Linear search ✓
d. Binary search

Question **10**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search can find multiple instances of the target element
c. Binary search works on unsorted lists
d. Binary search has a lower time complexity on large, sorted lists ✓

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By dividing the list length by two ✓
b. By comparing each element sequentially
c. By using a hash function
d. By starting from the first element

Question **12**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and unsorted
b. When the list is large and sorted
c. When the list is small and unsorted ✓
d. When the list is small and sorted

Question **13**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100
b. 500
c. 10 ✓
d. 1000

Question **14**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Interpolation search
b. Hash search
c. Linear search ✓
d. Binary search

Question **15**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Sequential search
b. Hash search
c. Linear search
d. Binary search ✓